A KNOCKOUT BLOW FOR LLMS?

EMPIRICAL LIMITS OF REASONING AND PLANNING IN LANGUAGE MODELS

Gary Marcus (Original Analysis)

Slides prepared by [Mei Nagappan]

Original article: https://garymarcus.substack.com/p/a-knockout-blow-for-llms

UNIVERSITY OF WATERLOO | FACULTY OF MATHEMATICS

Learning Goals

- Define “reasoning” in an operational, testable way
- Understand Apple's empirical evaluation of LLM reasoning
- Analyze Gary Marcus's critique of reasoning claims
- Connect these results to Rao Kambhampati's work on planning

Why This Debate Matters

- LLMs are used in domains requiring correctness, not plausibility
- Reasoning claims imply robustness under novelty and complexity
- Overestimating reasoning ability risks unsafe deployment

Operational Definition of Reasoning

- Consistent application of rules across problem instances
- Correct handling of increased problem depth or size
- Reliable extrapolation beyond the training distribution

The Apple Study (As Reported by Marcus)

- “Reasoning-augmented” LLMs evaluated (e.g., chain-of-thought)
- Used tasks with well-defined algorithmic solutions
- Varied task complexity systematically

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Key Empirical Result (Described)

- Performance is high on low-complexity instances
- Accuracy declines sharply as complexity increases
- Decline is monotonic and reproducible

Why This Pattern Matters

- Suggests absence of internal algorithm execution
- Indicates reliance on pattern matching
- Scaling input length does not resolve failures

Tower of Hanoi (Problem Description)

- Classical recursive planning problem
- Goal: move stacked disks under strict constraints
- Optimal solution follows a simple recursive algorithm

Why Tower of Hanoi Is Diagnostic

- Deterministic rules, no ambiguity
- Requires explicit state tracking
- Solution generalizes perfectly with recursion

Observed LLM Failures on Hanoi

- Correct for small numbers of disks
- Incorrect or inconsistent for larger instances
- Violates constraints or loses track of state

Marcus's Interpretation

- LLMs generate plausible sequences, not procedures
- No explicit variable binding
- No persistent internal state representation

Chain-of-Thought: Why It Misleads

- Produces fluent step-by-step text
- Steps may contradict earlier assumptions
- Reasoning text is not causally constraining

Generalization vs Extrapolation

- LLMs interpolate within training regimes
- Reasoning requires extrapolation to unseen depths
- Apple's results expose this gap

Hallucinated Reasoning

- Confidently stated incorrect intermediate steps
- Errors propagate unnoticed
- No internal consistency checks

Rao Kambhampati (As Cited by Marcus)

- Professor of Computer Science, ASU
- Expert in automated planning
- Critiques LLMs as unreliable planners

https://rakaposhi.eas.asu.edu

Kambhampati's Key Claim

- LLMs narrate plans rather than compute them
- They lack explicit state-transition models
- Planning competence is inferred incorrectly

Planning vs Plan Narration

- Planning requires constraint satisfaction
- Requires verification at each step
- LLMs lack grounded symbolic representations

Converging Evidence

- Apple: empirical degradation with complexity
- Marcus: conceptual critique of reasoning claims
- Kambhampati: planning-theoretic explanation

What This Critique Does NOT Say

- LLMs have no value
- Language modeling is trivial
- Progress is impossible

What It DOES Say

- LLMs are not general reasoners
- Fluency is not evidence of reasoning
- AGI claims require stronger proof

Implications for AI System Design

- Hybrid symbolic-neural systems
- External planners and verifiers
- LLMs as interfaces, not reasoning cores

Discussion

- Are LLMs fundamentally limited or just incomplete at least with respect to code generation?
- What empirical test in SE would convince you of reasoning?

References

- Marcus, G. https://garymarcus.substack.com/p/a-knockout-blow-for-llms
- Apple Research https://arxiv.org/abs/2506.06941
- Kambhampati, S. critique on reasoning https://arxiv.org/abs/2504.09762